Fiscal Year:	FY 2013	**Task Last Updated:**	FY 12/01/2012
PI Name:	Vickers, Zata Ph.D.		
Project Title:	Factors Contributing to Food Acceptability and Consumption, Mood, and Stress on Long-term Space Missions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	zvickers@umn.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	612-624-2257
Organization Name:	University of Minnesota		
PI Address 1:	Food Science and Nutrition		
PI Address 2:	1334 Eckles Ave		
PI Web Page:			
City:	Saint Paul	**State:**	MN
Zip Code:	55108-1038	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	02/15/2012	**End Date:**	02/14/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Douglas, Grace	**Contact Phone:**	
Contact Email:	grace.l.douglas@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mann, Traci (University of Minnesota) Redden, Joseph (University of Minnesota)		
Grant/Contract No.:	NNX12AE56G		
Performance Goal No.:			
Performance Goal Text:			

The three inter-related foci in this proposal explore factors that contribute to the long-term acceptability of a diet, in terms of satisfaction with and sufficient consumption of the foods, as well as associated moods and stress levels. The objective of our first focus is to examine behaviors related to preparing and consuming meals.

Understanding the relation between preparation effort and the acceptability and consumption of the prepared food, as well as effects on moods and stress, will inform decisions on the amount of time and types of activities crew members should allot to meal preparation. Understanding the relations between communal dining and special occasion meals and moods and stress will allow recommendations for dining behaviors that maximize positive moods and minimize stress.

To do this we will separately manipulate food preparation effort, food preparation creativity, and the number of people eating together. We will measure the fluctuation in participants' moods and stress levels using questionnaires and

Task Description:	physiological measurements. Meal and food acceptability will be measured using a variety of subjective rating scales. The objective of our second focus is to to assess whether comfort foods repair individuals' moods, and if so, whether comfort foods are more effective at repairing moods than control foods. Understanding the effects of specific foods on crewmembers' food consumption, stress, and moods will inform decisions on the quantities of such mood-improving foods that should be made available on long missions. Participants first complete an online questionnaire about their food preferences (session one), then they come to the lab for two separate sessions in which we induce a negative mood and then give subjects a food to eat. We measure their mood after the negative mood induction, and then again after they eat the food, to assess whether the food led to improvements in their mood. The objective of our third focus is to curb boredom with repeatedly eaten foods by increasing appreciation for variety in one's diet without changing the available food items. Because the most unpleasant mood/emotion associated with food is boredom, and because boredom is highly related with a perceived lack of variety, understanding the conditions influencing people's perceptions and satisfaction with the variety in a diet will enable the crafting of practices that positively affect the mood and satisfaction of crewmembers with the foods on extended missions. Subjects in this study participate in three phases: phase I identified a liked food for each participant; phase II satiated participants on that single food; and phase III tested whether recalling variety reduced satiety and increased subsequent intake. For the first two focuses, positive results in ground based studies will be tested in flight definition studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our investigation of the role of food preparation and the eating situation on meal satisfaction, consumption, moods, and stress is highly related to these same issues on Earth. Is the selection and preparation of food 'therapy' for relieving stress? Or does it add to stress? And does this depend on specific characteristics of the individual? Information in the popular press strongly suggests that some foods (e.g. comfort foods) can relieve stress and repair bad moods. Does this really work on Earth? Are 'comfort foods' more effective at reducing stress and repairing moods than other foods? Than other activities? Our investigation of perceived variety is relevant for Earth-based situations where the variety of food people have available is limited or for people on highly restricted diets.
Task Progress:	Overview. Our first year goals included making substantial progress on each of our three focuses. Each focus is briefly described below. For each Focus we describe the experiments we have conducted or are presently conducting. Focus 1. The role of food preparation and the eating situation on meal satisfaction, consumption, moods, and stress. Goals. The goals of this set of studies were to determine whether preparing food and/or having the ability to choose the ingredients in the food would improve mood and reduce stress. Methods. Participants came individually to our lab at dinner time. We measured stress (cortisol, heart rate, and blood pressure) and moods (profile of mood states) on arrival, then we induced stress and measured stress and moods again. Each subject was then assigned to one of four experimental conditions. In the prepare-choice condition participants prepared a meal (pasta + sauce + inclusions) themselves and had control over the selection of the meal components. In the prepare-no-choice condition subjects prepared their meal, but had no control over details of the menu. In the choice-no-prepare condition subjects had control over some details of the menu, but the meal was prepared by someone else. In the no-prepare-no-choice condition participants were simply provided with a meal prepared by someone else. Responses to a food preference questionnaire conducted before the stress induction were used to ensure that all participants were provided with foods they liked. Next Steps. We are continuing to analyze the data, examining effects of choice and preparation on measures of liking, consumption, etc. We are beginning to discuss plans for the flight studies with the goal of measuring astronauts' stress, moods, consumption, and meal satisfaction for a meal they eat alone, a meal they eat with others, and a meal they eat on a special day. Focus 2: The effects of specific food items on consumption, moods, and stress. Goals. The goal of this set of studies was to assess whether comfort foods repair individuals' moods, and if so, whether comfort foods are more effective at repairing moods than control foods. Methods. Participants first completed an online questionnaire about their food preferences, then they came to the lab for two separate sessions in which we induced a negative mood and then gave subjects a food to eat. We measured their mood after the negative mood induction, and then again after they ate the food, to assess whether the food led to improvements in their mood. Next Steps. We are continuing our data collection activities. We are also beginning to discuss plans to determine whether eating foods shown to be capable of improving moods in this focus 2 ground study will improve moods of astronauts in space when they are in negative moods. Focus 3: The role of perceived variety in meal satisfaction, consumption, and mood. Goals. The goal of these studies was to curb boredom with repeatedly eaten foods by increasing appreciation for variety in one's diet without changing the available food items. Methods. In phase I we identified a liked food for each participant; during phase II we satiated participants on that single food. During phase III participants in the experimental condition listed the various foods that they had eaten in the past week outside of the study, whereas participants in the control condition listed the musical artists they had listened to in

	the past week. We tested whether recalling variety in other foods they had eaten reduced satiety and increased subsequent intake. Overall, we have observed satiation in reported satisfaction/enjoyment of the food, but not always in the quantity eaten. Next Steps. We plan to continue the data collection for this study. On completion of that study we will proceed with our original plans for a ground study that tests the extent to which reminders of past variety help reduce satiation (monotony) with a diet over longer periods of time in natural settings.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Abstracts for Journals and Proceedings	Vickers ZM, Mann TL, Redden JP. "Factors contributing to food acceptability and consumption, mood, and stress on long-term space missions." Presented at the 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012. 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012. , Feb-2012